

VOLUME III - TECHNOLOGY / DOCUMENT 1 OF 4

Radar Research

Millimetre-Wave Radar as the Primary Non-Cooperative Sensing Layer

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ABSTRACT

This document examines millimetre-wave radar in depth, as the technology on which the residual-sensing layer of the proposed architecture depends. It covers FMCW operating principles and the resulting range, velocity and angular resolution; the atmospheric propagation physics that makes radar fog-tolerant; MIMO and 4D imaging radar; the specific challenges of a mining haul road - ground clutter on gradients, multipath from a ferrous ore body, mutual interference across a dense fleet, and radome fouling; radar odometry as a positioning aid; and a consolidated draft specification for procurement.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. Why Radar

The conclusion of Volume II Document 2 was that millimetre-wave radar is the only mature, affordable, rugged sensing modality whose physics survives a meteorological optical range of 3-5 m at the ranges the stopping-distance analysis demands. This document establishes what that means in engineering terms.

Four properties make radar the correct choice, and it is worth being explicit that **no single one of them would be sufficient on its own**:

- ? **Fog transparency.** At 77-79 GHz the wavelength is approximately 3.9 mm against fog droplets of 1-20 μm , a ratio of roughly 200-4000:1. Scattering is deep in the Rayleigh regime where the cross-section falls as the sixth power of particle diameter. The droplets are, for practical purposes, not there.
- ? **Direct velocity measurement.** Doppler processing yields radial velocity directly, at resolutions on the order of 0.1 m/s. This is not a convenience - closing rate is the quantity that determines time-to-collision, and measuring it directly rather than differentiating a noisy position estimate is what makes trajectory-aware alerting (requirement FR-19) practical.
- ? **Range.** Automotive millimetre-wave radar detects obstacles within approximately 250 m, against a typical LiDAR detection range within 150 m. The stopping-distance analysis requires 80 m with a goal of 120 m; radar provides margin.
- ? **Fouling tolerance.** This property is specific to mining and decisive. An optical window with a film of iron ore mud is opaque. A radome with the same film is attenuated. Radar degrades where optics fails.

2. FMCW Operating Principles

Automotive and industrial millimetre-wave radar is overwhelmingly frequency-modulated continuous wave. The transmitter emits a linear frequency sweep - a chirp - and the receiver mixes the return with the transmitted signal, producing a beat frequency proportional to target range.

2.1 Range measurement

For a chirp of bandwidth B swept over time T_c , a target at range R produces a beat frequency:

$$f_b = 2 R B / (c * T_c)$$

Range resolution follows from the bandwidth alone:

$$dR = c / (2 * B)$$

Bandwidth	Range resolution	Comment
250 MHz (24 GHz ISM)	0.60 m	Insufficient to separate a person from a nearby vehicle
1 GHz	0.15 m	Adequate
2 GHz	0.075 m	Good
4 GHz (79 GHz band)	0.0375 m	Excellent; separates closely spaced targets
5 GHz	0.030 m	Best available in the 76-81 GHz allocation

Table 2.1 - Range resolution as a function of sweep bandwidth. The wide bandwidth available in the 79 GHz band is a principal reason for preferring it over 24 GHz.

Maximum unambiguous range is set by the sampling rate and the intermediate-frequency bandwidth of the receiver, and in practice by the signal-to-noise ratio available at range, not by the chirp geometry.

2.2 Velocity measurement

A coherent sequence of chirps - a frame - allows a second FFT across chirps, extracting Doppler shift and hence radial velocity. For N_c chirps of duration T_c at wavelength λ :

$$\Delta v = \lambda / (2 N_c T_c) \text{ and } v_{\max} = \lambda / (4 * T_c)$$

At 77 GHz with a 50 μ s chirp and 128 chirps per frame, velocity resolution is on the order of 0.3 m/s and maximum unambiguous velocity approximately ± 19 m/s (68 km/h), comfortably covering haul truck speeds. Longer frames improve velocity resolution at the cost of frame rate; the requirement of PR-14 for 10 Hz update sets the ceiling.

Doppler is the key discriminator on a haul road. A stationary berm, a rock face and the road surface all return strong echoes but at zero Doppler relative to the ground. A moving truck, a person walking, a rolling rock all have non-zero Doppler. Doppler processing therefore separates the hazards from the environment before any classification logic runs - which is why radar performs well in a cluttered pit despite its modest angular resolution.

2.3 Angular measurement and MIMO

Azimuth and elevation are resolved by an antenna array. Angular resolution for a uniform linear array of N elements at half-wavelength spacing is approximately:

$$\Delta \theta \sim 2 / N \text{ radians (at boresight)}$$

MIMO - multiple-input multiple-output - synthesises a large virtual array from a smaller number of physical elements. With N_{tx} transmit and N_{rx} receive elements, a virtual array of $N_{tx} \times N_{rx}$ elements is formed, giving the angular resolution of the larger array from the hardware of the smaller.

Configuration	Virtual elements	Approx. azimuth resolution	Lateral resolution at 50 m
2 Tx x 4 Rx	8	~15 deg	~13 m
3 Tx x 4 Rx	12	~10 deg	~8.7 m
4 Tx x 4 Rx	16	~7 deg	~6.1 m
12 Tx x 16 Rx	192	~1.5 deg	~1.3 m
Cascaded imaging (48 Tx x 64 Rx class)	3072	~0.5 deg	~0.44 m

Table 2.2 - Angular resolution against MIMO configuration. Requirement PR-07 asks for discrimination of an in-lane from an adjacent-lane target at 50 m, implying roughly 3 deg or better.

Table 2.2 makes the case for imaging radar directly. A conventional 3x4 automotive corner radar cannot tell, at 50 m, whether a detected truck is in the operator's lane or the adjacent one. On a two-way haul road that distinction is the entire alerting decision. **The angular resolution requirement, more than the range requirement, is what drives the selection toward imaging-class radar.**

2.4 The resolution trade space

The designer trades among range resolution (bandwidth), velocity resolution (frame duration), angular resolution (array size), update rate, field of view and cost. For this application the priority order is:

- ? **Angular resolution** - drives lane discrimination, which drives alert correctness and hence the nuisance alert rate (HR-06, risk R3).
- ? **Range** - a hard gate from the stopping distance analysis.
- ? **Update rate** - 10 Hz minimum per PR-14.

? **Velocity resolution** - already adequate in standard configurations.

? **Range resolution** - already adequate above 1 GHz bandwidth.

3. Frequency Band Selection

Band	Allocation	Bandwidth	Range resolution	Aperture for given beamwidth	Assessment
24 GHz ISM	Narrowband ISM	250 MHz	0.60 m	Large	Rejected. Insufficient resolution; band being phased out for automotive use.
24 GHz UWB	Legacy UWB	~5 GHz	0.03 m	Large	Rejected. Regulatory phase-out; large antennas.
60 GHz	ISM, oxygen absorption peak	Wide	Fine	Moderate	Rejected for long range. The 60 GHz oxygen absorption line imposes roughly 15 dB/km additional loss - deliberately useful for short-range, but wrong for an 80-120 m requirement.
76-77 GHz	Automotive long range	1 GHz	0.15 m	Small	Suitable. Higher permitted EIRP for long-range forward sensing; the established long-range automotive band.
77-81 GHz	Automotive short/mid range	up to 4 GHz	0.04 m	Small	Preferred for imaging. Wide bandwidth, excellent range resolution, smaller aperture for a given beamwidth, mature automotive supply.
140 GHz and above	Research / emerging	Very wide	Very fine	Very small	Future option. Higher resolution and smaller apertures; immature supply chain and higher atmospheric loss.
300 GHz (low THz)	Research	Very wide	Very fine	Very small	Research only. Experimental work confirms object visibility through fire, smoke and steam, with higher frequencies offering reduced aperture for equivalent beamwidth - relevant to vehicle packaging - but the range above 100 GHz is insufficiently studied for production deployment.

Recommendation. The **76-81 GHz automotive allocation**, using 76-77 GHz configuration for the long-range forward sensor where permitted EIRP supports the 120 m goal, and 77-81 GHz for mid-range, imaging and corner sensors where bandwidth and aperture matter more. Indian spectrum authorisation for this band in a mining application must be confirmed early (requirement CR-07, risk R9).

4. Atmospheric Propagation and Fog

4.1 Scattering regime

The scattering regime is set by the size parameter $x = \pi \cdot d / \lambda$. For a 10 μm droplet at 77 GHz ($\lambda = 3.9 \text{ mm}$), x is approximately 0.008 - deep in the Rayleigh regime, where the scattering cross-section scales approximately as d^6 / λ^4 . The same droplet at 905 nm gives x of approximately 35, firmly in

the Mie regime where scattering is strong and efficiency is near its maximum.

This is a difference of regime, not of degree. It is the physical basis for the entire architecture.

4.2 Published attenuation data

Condition	77 GHz radar	905 nm LiDAR	Source character
Heavy rain, 25 mm/hr	~10 dB/km	~35 dB/km under comparable low visibility	Published comparative survey
Heavy fog	Modest; detection range ~260 m reported at 20 dB SNR threshold for a 77 GHz configuration	Fails beyond ~70 m in dense fog per fusion simulation studies	Simulation and survey literature
Mild fog	Detection range ~460 m reported	Substantially degraded	Simulation
Snowfall	Attenuation rises with snow rate; wet snow worse due to higher water absorption and larger flakes	Severe	Measurement
Fire, smoke, steam	79 GHz and 300 GHz confirmed to render objects visible; attenuation measured and compared with theory	n/a	Experimental

Table 4.1 - Published attenuation and range behaviour. Values are drawn from the open literature under the stated conditions and are indicative; site measurement at Bailadila remains required per requirement ER-10 and AR-02.

An important caveat for honest engineering: **the published fog figures are largely simulation-derived or from controlled fog chambers.** Monsoon orographic cloud on an ore ridge, with liquid water content and droplet size distribution that may differ from the modelled cases, and with heavy rain frequently superimposed, is not the same medium. The literature establishes that radar is the right modality by a wide margin; it does not establish the exact detection range at Bailadila. That is what the validation plan in Section 12 exists to determine.

4.3 Effect on the radar range equation

The monostatic radar range equation gives received power:

$$P_r = (P_t G^2 \lambda^2 \sigma) / ((4\pi)^3 R^4 L)$$

where σ is the target radar cross-section and L the total loss including atmospheric attenuation. Atmospheric loss over a two-way path of range R at specific attenuation α dB/km is $2 \alpha R / 1000$ dB.

At an assumed heavy-fog specific attenuation of, say, 5 dB/km at 77 GHz, the two-way loss at 120 m is approximately **1.2 dB** - a 25 per cent reduction in received power, which is recovered by a modest increase in integration time or transmit power. The R^4 term dominates the link budget overwhelmingly; fog is a second-order correction.

This calculation is the crux of the argument. For an optical sensor at MOR 5 m, the two-way loss at 120 m is astronomically large - the signal is extinguished many times over within the first few metres. For radar it is roughly one decibel. **Fog is a design consideration for radar and a design impossibility for optics.**

5. 4D Imaging Radar

4D imaging radar extends conventional automotive radar by adding elevation as a resolved dimension, producing range, azimuth, elevation and Doppler - hence "4D" - and outputting a sparse point cloud rather than

a short target list.

Why elevation matters on a haul road. A conventional 2D automotive radar cannot distinguish a target in the travel path from one above or below it. On a flat highway this is tolerable. On a haul road it is not, for three specific reasons:

- ? **Gradients.** On a 1-in-16 descent, the road surface ahead falls away from the sensor boresight. Without elevation resolution, the radar cannot separate returns from the road surface, from a berm at the outside of a curve, and from a genuine in-path obstruction.
- ? **Switchbacks.** On a hairpin, the road above and the road below are within the same azimuth-range cell. A truck on the level above is not a hazard; an equivalent return at the same range and azimuth on the operator's own level is. Elevation resolves the ambiguity.
- ? **Overhead structures.** Conveyor gantries, bridges and highwall features generate returns that a 2D radar must reject by heuristics and an elevation-resolving radar rejects geometrically.

Sector adoption. Hexagon has introduced 4D radar support into its mining collision avoidance system, indicating that the mining sector has independently reached the same conclusion.

Consequences for the design. Imaging radar produces substantially more data - thousands of points per frame rather than tens of targets - requiring more compute and a point-cloud processing chain closer to LiDAR practice than to classical radar practice. This is manageable on an automotive-grade SoC with an NPU, and the algorithmic tooling is well developed.

6. Haul Road Specific Challenges

6.1 Ground clutter on gradients

The road surface itself is a large, rough, strongly reflecting extended target. On a descent, the geometry between sensor boresight and surface changes continuously with vehicle pitch, so a fixed elevation mask is inadequate.

Mitigations.

- ? Elevation-resolving radar with a ground plane estimated dynamically from the vehicle's own pitch (from the IMU) and from the digital road model's known gradient ahead.
- ? Doppler-based static clutter rejection: the road surface returns at a Doppler consistent with the vehicle's own ground speed, which is known precisely from L1, allowing coherent removal.
- ? Constant false alarm rate detection adapted to the non-homogeneous clutter of a bermed, graded road rather than the homogeneous assumption of classical CFAR.
- ? Use of the digital road model as a spatial prior: returns coincident with known static geometry are down-weighted, and returns in the known travel corridor are escalated. **This is a direct and valuable coupling between the map layer (L2) and the sensing layer (L4).**

6.2 Multipath in a ferrous pit environment

The Bailadila ore body is high-grade haematite - among the highest grade in the world - surrounded by benched rock faces and populated by very large steel vehicle bodies. This environment generates strong specular reflections and multipath, which can produce ghost targets at plausible ranges and azimuths.

Mitigations.

- ? Track-level rejection: ghosts typically exhibit kinematics inconsistent with a physical vehicle - impossible accelerations, trajectories through solid geometry - and are rejected by validating tracks against the road

model.

- ? Cross-validation against the cooperative layer: a strong radar return at a position where no cooperative broadcast originates is either a genuine non-cooperative hazard or a ghost, and its persistence and kinematic plausibility discriminate.
- ? Elevation resolution rejects a substantial class of ground-bounce multipath.
- ? **On-site measurement is essential.** Requirement ER-10 exists specifically for this. Multipath behaviour in a high-grade haematite pit cannot be predicted from generic automotive or mining test data.

6.3 Mutual interference

With dozens of vehicles each carrying several radars operating in the same band in a confined pit, mutual interference is a first-order concern, manifesting as elevated noise floor or as false detections.

Mitigations.

- ? Chirp randomisation: varying slope, start frequency and timing between frames so that interference decorrelates and appears as noise rather than as a coherent false target.
- ? Interference detection and mitigation in the signal processing chain - detection of corrupted chirps and their exclusion or repair, which is standard in modern automotive radar transceivers.
- ? Coordinated time or frequency allocation via the V2V link, using the cooperative layer to schedule radar resources - technically attractive and worth evaluating, though it introduces a dependency between layers that principle P3 cautions against and should therefore be an optimisation, never a requirement for safe operation.
- ? Design-phase modelling of the worst-case density defined in CR-03 (60 vehicles within mutual range).

6.4 Radome fouling

A radome accumulating iron ore mud attenuates, detunes and can introduce phase distortion across the aperture, degrading angular accuracy before it degrades detection.

Mitigations.

- ? Hydrophobic radome coatings and sheltered mounting geometry, avoiding upward-facing or spray-exposed surfaces.
- ? Continuous **blockage self-detection**: modern automotive radar transceivers can estimate radome attenuation from the return statistics, and this must be exposed to the fusion layer to satisfy requirement FR-17.
- ? Declared degraded mode with visible effective-range reduction on the operator display, rather than silent performance loss - satisfying HR-11 and mitigating failure mode F3.
- ? Routine cleaning integrated into the pre-shift checklist (RR-03), and empirical measurement of fouling rates during Phase 0 so that the cleaning interval is evidence-based.

Radome fouling is where the radar advantage is most often overstated and must be handled honestly. Radar tolerates fouling far better than optics, but it does not ignore it. A heavily caked radome can lose several decibels, which at the R^4 dependence of the range equation is a material reduction in detection range. The system must know this and say so.

6.5 Vibration and boresight stability

HEMM vibration can shift sensor boresight, and an angular error of a degree at 80 m is a lateral error of 1.4 m - enough to place an in-lane target in the adjacent lane, or the reverse.

Mitigations. Rigid mounting to structurally stiff points; documented installation procedure with an alignment step; **online extrinsic calibration monitoring** (requirement RR-06) comparing radar-derived static geometry against the digital road model and the vehicle's own motion, with automated drift alerting.

7. Target Characteristics and Detection Budget

Radar cross-section governs detectability. Indicative values at millimetre-wave frequencies:

Target	Indicative RCS	Detection significance
Haul truck, rear aspect	Very large, tens of m ²	Easily detected well beyond requirement
Light vehicle	Several m ²	Comfortably detected at PR-02 range of 50 m
Person, unaugmented	~0.5-1 m ² , highly aspect-dependent	The critical case. Meeting PR-03 at 25 m requires adequate sensitivity and careful clutter handling
Person with retro-reflective PPE	Enhanced	PPE with radar-reflective elements would materially improve detection; worth investigating as a low-cost complementary measure
Static rock, 0.5 m	Low to moderate, geometry-dependent	PR-04 case; hardest non-cooperative target; benefits from imaging resolution
Berm	Large extended target	Useful as a road-edge reference , not only as clutter - see Section 8
Standing water / washout	Very low; specular reflection away from sensor	Effectively undetectable by radar. Must be handled by the map layer and by control room hazard markers (FR-22)

Table 7.1 - Indicative target characteristics. Values are order-of-magnitude and require measurement on site.

Two entries in this table are worth drawing out.

The washout case is a genuine architectural limitation and must be stated plainly. A flooded or scoured section of haul road presents a smooth surface that reflects specularly away from the radar, returning little energy. Radar will not reliably detect it. This hazard - which is specifically a monsoon hazard - must be covered by the road model, by control room hazard markers placed by road maintenance crews, and by procedure. No sensing layer in the proposed architecture solves it. Recognising this is more useful than assuming radar covers everything.

The berm is an asset, not only clutter. A well-formed berm is a large, continuous, geometrically regular extended target running parallel to the road on both sides. It is an excellent radar reference for lateral position and provides an independent cross-check on the GNSS/INS lane solution - valuable during GNSS outage.

8. Radar Odometry and Localisation

Radar can contribute to positioning as well as detection, which is valuable given the GNSS shadowing risk (R2).

- ? **Radar odometry.** Successive radar frames are registered against each other to estimate ego-motion, analogous to LiDAR odometry but functioning in conditions where LiDAR does not. Research demonstrates radar-aided SLAM in low-visibility fog environments, including work using a mechanical pivoting radar with only a 15 m measurement range whose fog penetration was sufficient for landmark searching where LiDAR failed.
- ? **Radar landmark matching.** Persistent strong scatterers - berm corners, light masts, plant structures, rock face features - can form a radar landmark map against which the vehicle localises. This is fog-immune and

infrastructure-free.

- ? **Berm tracking for lateral position.** As noted above, direct measurement of distance to the berms on both sides gives an independent lateral position within the running surface, directly cross-checking the primary lane solution.

Recommendation. Radar odometry and berm tracking should be implemented as **aiding inputs to the L1 positioning filter**, not as a standalone solution. Their value is highest exactly when GNSS is least available - in deep pit sections and against high rock faces - which is the complementarity the architecture needs. This is a software capability built on hardware already present for detection, so the marginal cost is development effort only.

9. Sensor Placement and Coverage Design

Indicative configuration for a large rear-dump truck. Final configuration depends on machine geometry and must be validated per machine class.

Position	Type	Coverage	Purpose
Front centre, high	Long-range imaging radar, 76-77 GHz	+/- 15 deg azimuth, 120 m+	Primary forward hazard detection; PR-01, PR-02
Front centre, low	Mid-range imaging radar, 77-81 GHz	+/- 60 deg azimuth, 60 m	Wide forward coverage on curves; PR-03, PR-04
Front corners (2)	Short/mid-range radar	+/- 75 deg, 40 m	Curve entry, junction approach, near-field sides
Rear corners (2)	Short/mid-range radar	+/- 75 deg, 40 m	Reversing, tipping approach, rear blind zone; FR-13
Rear centre	Short-range radar	+/- 60 deg, 25 m	Rear blind zone; complements the DGMS-mandated rear camera
Sides (2)	Short-range radar	+/- 60 deg, 20 m	Lateral blind zones, personnel protection

Design notes.

- ? On a switchback the forward long-range sensor points at a rock face for much of the curve. **Wide-azimuth mid-range coverage matters more than peak range on curved sections**, and the digital road model should be used to steer attention - known curvature ahead tells the fusion layer where the road goes, so that returns in that direction are prioritised even when they are far off boresight.
- ? Mounting heights must account for the very high cab position of a large rear-dump truck; a sensor mounted at cab height on a 100 t truck looks down at a light vehicle at close range, which is an elevation geometry that a 2D radar handles poorly and an imaging radar handles correctly.
- ? All mountings must be sheltered from direct spray and avoid upward-facing surfaces (Section 6.4).

10. Signal Processing Chain

Stage	Function	Notes for this application
ADC and range FFT	Beat frequency to range bins	Standard
Doppler FFT	Range-Doppler map	Static clutter identified against known ego-velocity from L1
Interference detection and mitigation	Identify and repair corrupted chirps	Essential at the fleet densities of CR-03

Stage	Function	Notes for this application
Angle estimation (MIMO beamforming or super-resolution)	Azimuth and elevation	Determines lane discrimination; the critical stage for HR-06
CFAR detection	Adaptive thresholding	Must be adapted to non-homogeneous haul road clutter
Point cloud formation	4D detections	Imaging radar output
Ground and static clutter removal	Remove road surface, berms, rock faces	Uses ego-velocity, IMU pitch and the digital road model as priors
Clustering	Group detections into objects	DBSCAN-class methods, tuned for sparse radar point clouds
Tracking	Multi-target tracking with identity, velocity, predicted trajectory	Extended Kalman or IMM filters; satisfies FR-15
Classification	Assign target class with confidence	Radar micro-Doppler is informative for pedestrian discrimination
Fusion with cooperative layer	Associate and resolve duplicates	Satisfies FR-16
Self-diagnosis	Blockage, interference level, calibration drift	Satisfies FR-17, RR-06

Micro-Doppler note. The gait of a walking person produces a characteristic micro-Doppler signature - limb motion modulating the return - that distinguishes a pedestrian from a rigid object of similar cross-section. This is well documented in the radar literature and is the most reliable radar-only means of pedestrian classification. It is directly relevant to PR-03 and to the highest-consequence hazard class.

11. Draft Radar Subsystem Specification

Parameter	Requirement	Traces to
Frequency band	76-81 GHz, type-approved for use in India	CR-07, GR-04
Sweep bandwidth	≥ 1 GHz long-range; ≥ 2 GHz mid and short-range	PR-06
Range, vehicle target	≥ 120 m detection, ≥ 80 m with full track quality	PR-01
Range, person target	≥ 30 m detection, ≥ 25 m with full track quality	PR-03
Range accuracy	Better than ± 0.5 m or 2% of range	PR-06
Range resolution	≤ 0.3 m	PR-06
Velocity resolution	≤ 0.3 m/s	PR-06
Velocity range	± 30 m/s minimum	Operational
Azimuth resolution	≤ 3 deg at boresight; goal ≤ 1.5 deg	PR-07
Azimuth field of view	≥ 100 deg for mid-range; ≥ 30 deg long-range	Coverage design
Elevation resolution	≤ 5 deg; elevation resolution mandatory	Section 6.1
Update rate	≥ 20 Hz per sensor	PR-14 with margin
Probability of detection	≥ 0.95 per target per second in range	PR-05
Interference mitigation	Chirp randomisation and in-chain interference detection mandatory	Section 6.3
Blockage detection	Continuous radome attenuation estimate exposed to host	FR-17

Parameter	Requirement	Traces to
Ingress protection	IP67 minimum; IP69K where pressure-washed	ER-01
Operating temperature	-40 to +85 deg C at the sensor	ER-02
Vibration and shock	Qualified to HEMM service levels	ER-04
Interface	Automotive Ethernet or CAN FD; open documented protocol	IR-01
Output	4D point cloud plus tracked object list with confidence	FR-14, FR-15
Fog performance	Detection performance stated and demonstrated against measured MOR down to 5 m	AR-02, AR-03 - the differentiating requirement

12. Validation Plan

Radar is the layer carrying the largest single technical uncertainty in the architecture (risk R1). Validation should be front-loaded.

Stage	Activity	Timing	Success criterion
V1	Bench characterisation of candidate sensors against standardised targets in clear conditions	Phase 0	Meets range, resolution and accuracy specification
V2	Controlled fog chamber or artificial fog testing at measured MOR	Phase 0	Detection range degradation quantified at MOR 5, 10, 20, 50 m
V3	On-site Bailadila installation on an instrumented vehicle with co-located calibrated visibility meter, logging continuously through a monsoon season	Phase 0/1, monsoon-gated	Detection performance recorded against measured MOR across the natural range of conditions
V4	Multipath and ghost characterisation in the pit environment, against high-grade ore faces	Phase 0/1	Ghost rate quantified; rejection logic validated
V5	Interference testing at representative fleet density	Phase 1/2	No material degradation at CR-03 density of 60 vehicles
V6	Fouling rate measurement with periodic controlled cleaning	Phase 0, monsoon	Cleaning interval established empirically; blockage detection validated against measured attenuation
V7	Boresight stability over an extended service period	Phase 2	Drift within tolerance; online calibration monitor validated
V8	Full-system detection acceptance per AR-03	Phase 2, monsoon-gated	PR-01 to PR-07 demonstrated at measured MOR spanning V2, V3, V4

V3 is the single highest-value activity in the entire research programme and has the longest lead time, because it can only be conducted during a monsoon. It requires one instrumented vehicle, a candidate radar, a calibrated visibility meter and a data logger - a small fraction of the programme cost. It should be initiated at the earliest possible monsoon window, ahead of and independently of any procurement decision, because its results determine whether the architecture's central assumption holds at Bailadila and because a missed window costs a full year (risk R10).